

External finance & the macro link

Week 38 — when one firm's balance sheet becomes everybody's problem

Advanced Corporate Finance (EBM232A05)

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Two rows in the ledger, and one piece of fine print we have not paid for:

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	leverage $\downarrow$	flexibility survey (Graham and Harvey, 2001)
38	Costly external finance	? today	—

Last week's model *assumed* outside finance was costly. Today that assumption becomes the subject — and it turns out to be the friction that scales up from one firm to the whole economy.

Today

- The debt we owe from last week: why could that firm **only** borrow?
- The benchmark you own: investment and financing are **separate** — balance sheets should not matter.
- The puzzle: they visibly **do** matter — and why the obvious test of that is contested.
- The twist: a collateral limit turns net worth into a **multiplier**, $dl/dw = 1 + \lambda$.
- The pay-off: do it to **every** firm at once, and the micro friction *is* the macro accelerator.
- Evidence: the collateral channel, and the pecking-order horse race we promised you.

One friction — **costly external finance** — one new object: the firm's own *net worth*.

The debt we owe from last week

Last week's fine print

The flexibility model rested on one assumption we waved through: at $t=1$ the project could be financed **only with debt**. External equity was “prohibitively costly”.

- That assumption is **load-bearing**. In the problem set you replaced it: let the firm issue equity at issuance cost κ per euro, and the threshold becomes $\tau \geq p \kappa$.
- Let $\kappa \rightarrow 0$ and the flexibility motive **disappears**: a firm that can raise outside money for free never needs to preserve capacity.

Flexibility has value *because outside finance is costly*. So: how costly, why, and what does it do?

**The puzzle: balance sheets should
not matter**

The benchmark you own: separation

From the bachelor (Fisher separation, and MM's investment corollary): a firm takes a project if it is **positive-NPV**, full stop. *Who* funds it is a separate question with no effect on *whether* it happens.

$$\max_I \Pi(I) - (1+r)I \quad \implies \quad \Pi'(I^*) = 1+r,$$

and I^* does not depend on the firm's cash, its net worth, or last year's profits.

- In symbols: $\frac{\partial I^*}{\partial W} = 0$. **The balance sheet is irrelevant to the investment decision.**
- Perfect capital markets do the work: any positive-NPV shortfall is filled by outside investors at the fair rate.

This is the null. Everything today is a measurement of how badly it fails.

The puzzle: investment tracks cash flow

Fazzari et al. (1988) regress investment on Tobin's q — market value over replacement cost, a stand-in for opportunities — *and* on cash flow. Under separation the cash-flow coefficient should be **zero**. It is large, and largest for the firms that look most constrained.

- Read literally: **firms invest out of what they have**, not out of what is worth doing.
- But Kaplan and Zingales (1997) object, sharply: re-reading the same firms' filings, they find the *least* constrained ones show the *highest* sensitivity.
- The problem is not the fact — it is that q is a **noisy proxy** for opportunities, so cash flow may just be carrying news about the future.

A live methodological fight, and a good one to have read: the assigned Kaplan and Zingales (1997) *is* the critique.

The cleaner test: look at cash, not investment

Almeida et al. (2004) change the dependent variable. Instead of asking how firms *spend* a windfall, ask how much of it they **keep**.

- A firm that expects to be short of funds tomorrow should **hoard** some of today's cash flow.
- An unconstrained firm has no reason to: it can always raise what it needs.
- Prediction: the **cash-flow sensitivity of cash** is positive for constrained firms and zero for the rest.

Constraints are real. Now build the model that says what they *do*.

And that is what they find. Note what makes it clean: the *sign* of the prediction differs across groups, so a mis-measured q cannot generate it.

The twist: net worth as a multiplier

One new ingredient: only part of the return is pledgeable

Keep the project. Add one friction: an outside investor can only be promised what a court can **verify or seize**. So lending is capped by what the borrower already owns.

- **Budget**: the firm funds investment from its own net worth w plus borrowing b , so $I = w + b$.
- **Collateral limit**: $b \leq \lambda w$. One euro of the firm's own money levers in at most λ euro of outside money.
- λ rises with **tangibility**, creditor protection, and transparency. $\lambda \rightarrow \infty$ is MM.

Compare last week: capacity was a fixed number $\bar{D}$. The one new idea is that capacity is **proportional to what the firm owns**. Where λ comes from — which promises are credible — is contract theory: EBM008D05's lecture, not ours (Holmström and Tirole, 1997).

The constrained firm's investment

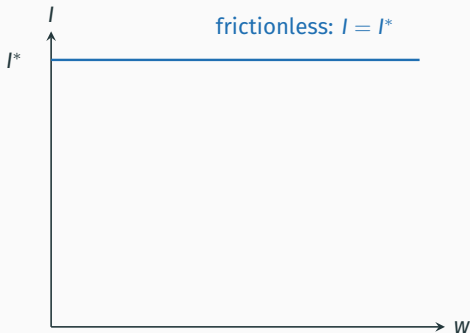
Substitute the limit into the budget:

$$I = w + b \leq w + \lambda w = (1 + \lambda) w, \quad \text{so} \quad I = \min\{I^*, (1 + \lambda) w\}.$$

- **Rich enough** $((1 + \lambda)w \geq I^*)$: the constraint does not bind — back in the benchmark.
- **Not rich enough**: the firm invests everything it can raise, and the project's NPV never enters the decision. The switch happens at $w^* = I^*/(1 + \lambda)$.

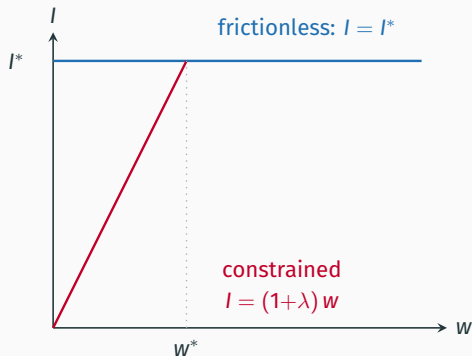
What is left out — ask it of every model: one period, no equity issue, a hard limit rather than a rising cost of funds. Softening any of them blurs the kink; none removes the multiplier.

Investment against net worth



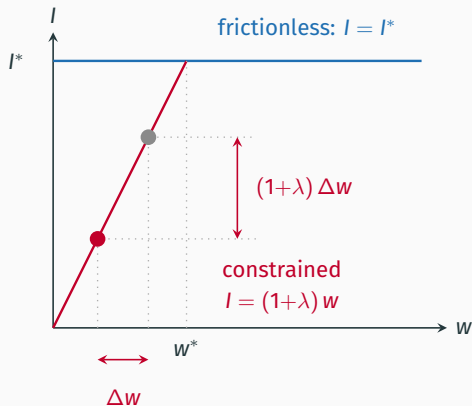
The benchmark, drawn: net worth on the horizontal axis, investment on the vertical. Separation says the line is **flat** — a poorer firm invests exactly as much as a rich one.

Investment against net worth



Now impose the collateral limit. Left of w^* the firm cannot reach I^* : investment rides a ray of **slope $1 + \lambda$** — steeper than one, because each euro it owns brings λ more with it.

Investment against net worth



Take a euro of net worth away from a firm on that ray. Investment does not fall by a euro — it falls by $(1 + \lambda)$ euro. The friction is not a tax on investment; it is a **lever** on it.

Net worth as a multiplier

Differentiate the constrained branch. Against a benchmark in which the answer is exactly zero:

$$\frac{dI}{dw} = \underbrace{1}_{\text{the euro itself}} + \underbrace{\lambda}_{\text{the borrowing it supported}} > 1 \quad \text{versus} \quad \frac{\partial I^*}{\partial w} = 0.$$

- A shock to the **balance sheet** is transmitted to the **real** economy, magnified by $1 + \lambda$.
- And note the sign of the comparative static that follows: the better the collateral ($\lambda \uparrow$), the **more** investment — but also the **more amplification**.

Everything else today is this one derivative, taken seriously.

Put in numbers

Net worth $w = 100$, a euro of borrowing per euro of its own ($\lambda = 1$), a project worth funding up to $I^* = 200$. A recession destroys $\Delta w = 20$.

	Net worth w	Borrowing b	Investment I
Before	100	100	200
After the shock	80	80	160 ↓
Frictionless, after	80	120 ↑	200

- €20 of **balance sheet** destroyed costs €40 of **investment**. The last row is the counterfactual: with no limit the firm simply borrows **more**.
- Turn the dial, across *constrained* firms: the same €20 costs €60 at $\lambda = 2$ and €20 at $\lambda = 0$ — better collateral buys **more investment and more amplification**.

Who is most exposed?

Read the comparative statics off $I = \min\{I^*, (1 + \lambda)w\}$ — and they are the cross-section:

Firm with...	...and so	Sensitivity to w
thin net worth (young, small)	sits on the ray	$1 + \lambda$
deep net worth (mature, cash-rich)	sits on the flat part	0
tangible assets ($\lambda \uparrow$)	borrow more per euro	$\uparrow$ larger
intangible, opaque assets	borrow little per euro	$\downarrow$ smaller

Two pay-offs. First, sensitivity is **not monotone in “constrainedness”** — a firm right at w^* jumps from $1 + \lambda$ to 0 — which is Kaplan and Zingales (1997)’s point, now as a *prediction* rather than an objection. Second, **tangibility raises the multiplier**: a testable sign we come back to.

From one balance sheet to the business cycle

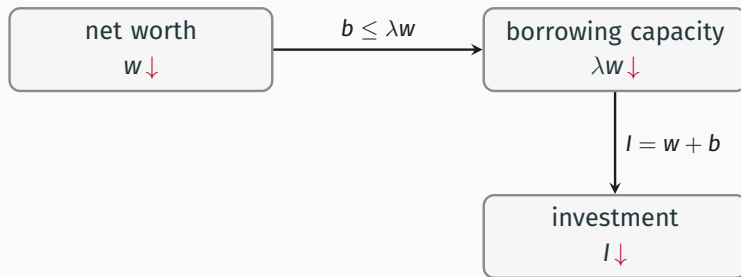
Now do it to everyone at once

So far, one firm and a given λ . But recessions do not destroy one balance sheet — they destroy **all of them, together**, and the collateral is a *traded asset*.

- Every constrained firm cuts investment by $(1 + \lambda)\Delta w$. Aggregate demand for capital — machines, land, real estate — **falls**.
- So its **price** falls — call it q , the price of the asset itself, not the ratio from two sections ago. And q is what the collateral is **worth**.
- So net worth is marked to market and falls **again**; capacity falls again; investment falls again.

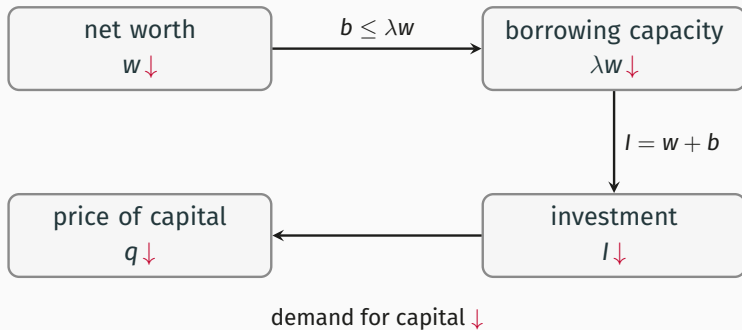
The firm-level multiplier was a number. Aggregated, it becomes a **loop**.

The loop



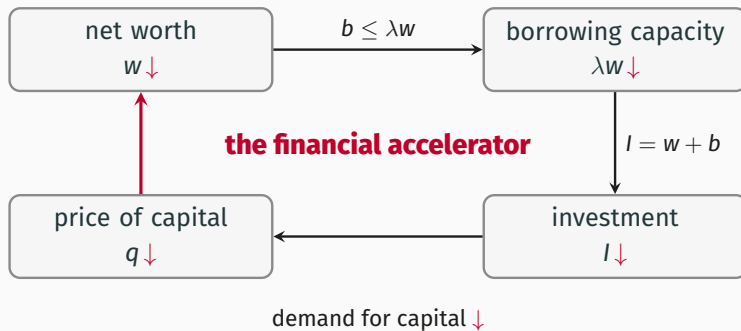
The chain we just derived, for one firm: net worth falls, the collateral limit $b \leq \lambda w$ bites, and the budget $I = w + b$ turns a euro of lost net worth into $(1 + \lambda)$ euro of lost investment.

The loop



Now every firm does it at once, so the **price of capital falls**. Name what changed: for one firm q was a **parameter**; once all firms move together it is a **variable**. That is the difference between a constrained-firm story and a macro-finance one.

The loop



And the collateral is that asset: it is **marked to market**, so the price fall lands back on net worth and the whole thing goes round again. **The micro friction is the macro accelerator** (Kiyotaki and Moore, 1997; Bernanke and Gertler, 1989).

That loop is what the literature calls the **financial accelerator**. Kiyotaki and Moore (1997) put it into a general-equilibrium model; two results are worth carrying:

- **Amplification**: a small, one-off shock produces a large fall in output — $(1 + \lambda)$ compounded through the price of collateral.
- **Persistence**: the shock outlives itself. Low investment today means less net worth tomorrow, so the economy **crawls** back rather than snapping back.

Kiyotaki and Moore (1997) — a **quantity** limit tied to collateral value: the *asset price* amplifies.

Bernanke and Gertler (1989) — a monitoring-cost **premium**: the cost of *funds* does, falling as net worth rises.

Complements, not rivals — both make **net worth** the state variable. That is the sense in which corporate finance is macroeconomics.

Evidence

The collateral channel

The model says: raise a firm's collateral value, hold its opportunities fixed, and investment rises. Chaney et al. (2012) find variation of that kind — local **real-estate** prices, which move firms' buildings for reasons unrelated to their projects.

- A euro more collateral buys roughly **six cents** more investment: λ is not a theorist's device.
- Almeida and Campello (2007) test the other sign: investment-cash-flow sensitivity **rises with tangibility**, among constrained firms only — the “credit multiplier”.

Read the identification, not just the coefficient: what makes Chaney et al. (2012) persuasive is that the variation in collateral comes from the **housing market**, not from the firm.

The horse race we promised you

Week 37 left profitability's negative sign on leverage unexplained. The **pecking order** explains it: if outside finance is costly because of asymmetric information (Myers and Majluf, 1984), firms fund internally first, then debt, equity last — profitable firms borrow less *because* they need less.

Test	Prediction	Finding
Shyam-Sunder and Myers (1999)	Δ debt tracks the deficit 1-for-1	coefficient ≈ 0.75
Frank and Goyal (2003)	same, broad panel	far below 1
Frank and Goyal (2009)	profitability $\rightarrow$ leverage $\downarrow$	robust

The strict *ordering* loses — equity tracks the deficit too, worst of all in the small, young firms where adverse selection should bite hardest. What survives is the premise: **external finance is costly**. The Myers–Majluf derivation itself is EBM008D05's.

Ledger & wrap

A financing friction does not just make investment **smaller**. It makes investment a **function of the balance sheet** — and a function with slope $1 + \lambda$ rather than 0. That is what turns a corporate-finance friction into a macroeconomic mechanism.

Two things to carry forward:

- “Identify the friction” now includes **aggregation**: the same constraint, imposed on everyone simultaneously, behaves differently than it does on one firm.
- The quantity that does the work is **net worth** — the one variable the frictionless benchmark insists cannot matter.

The frictions ledger — Week 38

Three rows now. Bring this back next week.

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36	Personal taxes (Miller)	value $\uparrow / \downarrow$	net benefit (Korteweg, 2010)
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38	Costly external finance	investment $\downarrow$	cash hoarding (Almeida et al., 2004)

And this week's row carries something the others do not: a **size**. Investment falls by $(1+\lambda)\times$ the shock, which is why this friction leaves the firm and shows up in the aggregate.

The recurring question, every lecture: **which friction, and in which direction?**

What is new here — and what is not

You already own:

- NPV, and separation of investment from financing
- that small firms find funding hard
- collateral as something lenders want

This week adds:

- net worth as a **multiplier**, $1 + \lambda$
- the **aggregation** step: micro friction
→ macro accelerator
- constraints as something you can **measure** and **identify**

Same machinery — a budget constraint and a derivative — pointed at a question the bachelor never asks: what happens when *every* firm is constrained at once?

Reading, and this week

- **This week:** Stein (2003) (the survey) or Kaplan and Zingales (1997) (the critique — shorter, and more fun to argue with).
- **Evidence to keep:** Almeida et al. (2004) (cash-flow sensitivity of cash) and Chaney et al. (2012) (the collateral channel).
- **Tutorial (Wed):** the project's **scoping memo** is due — it is a gate — plus the panel and regression methods your replication will need.
- **Next week:** we have treated “the firm” as one decision-maker. Week 39 splits it: the **manager** and the shareholders want different things.

A constraint on one balance sheet is a nuisance. The same constraint on every balance sheet is a recession.

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